

Unified Multi-Environment Cyber Threat Detection & Response Platform

Multi-Cloud Telemetry Ingestion • MITRE ATT&CK Matrix • ML Anomaly Engine • Google Gemini 2.0 Flash • SHA-256 Audit Ledger

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OBJECTIVE

Develop and deploy an **autonomous multi-environment Security Operations Center (SOC)** platform that unifies cloud and on-premise log ingestion, detects advanced threats via MITRE ATT&CK and ML anomaly scoring, maintains tamper-proof SHA-256 forensic audit trails, and provides real-time threat intelligence through an integrated **Google Gemini 2.0 Flash AI Analyst**.

DETAILED SYSTEM ARCHITECTURE & WORKFLOW PIPELINE



STAGE 1

Log Ingestion

AWS, Azure & Linux Syslog live collectors



STAGE 2

Normalization

Elastic ECS 1.12 schema standardization



STAGE 3

ML Risk Engine

Anomaly scoring (0-100) + Sigma rules



STAGE 4

MITRE Matrix

Automated TTP kill-chain categorization



STAGE 5

Audit Ledger

SHA-256 cryptographic chain proof



STAGE 6

AI Response

Gemini 2.0 Flash SOC chat & reports

Student View Point (In Own Words)

"As a student security operator, working with raw cloud logs used to feel overwhelming. This platform gave me an intuitive experience: I register my account, pick my profile from the login screen, upload messy multi-cloud logs, and immediately see clear threat classifications instead of endless text. Whenever a critical alert triggers, I simply ask the integrated Gemini 2.0 Flash AI to explain the attacker's motive and give me the exact containment commands. The SHA-256 audit chain guarantees my forensic evidence is mathematically unalterable."



1. One-Click Login
Recognizes registered users



2. Drag-Drop Logs
Instant ECS 1.12 parsing



3. Ask Gemini AI
Instant TTP breakdown



4. Download Proof
Signed forensic reports

Output View Point (Deliverables)

SOC Dashboard

Real-Time Threat Telemetry: Live metrics showing total ingested events, active threats, open incidents, and MTTD (<5 min).

MITRE Matrix

Automated Kill-Chain Mapping: Visual matrix highlighting active technique badges (T1078, T1059, T1486) with severity risk gauge.

Gemini AI Console

Conversational AI Analyst: Multi-turn LLM chat providing attack narrative, remediation code, and Sigma detection rules.

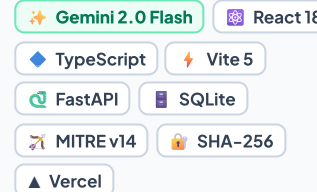
Forensic Ledger

Cryptographic Audit Proof: Downloadable PDF/CSV reports with SHA-256 hash chains verifying zero tampering.

Features & Tech Stack

- ✓ **Cross-Cloud Telemetry:** AWS, Azure, Linux
- ✓ **ML Anomaly Engine:** 0-100 risk scoring
- ✓ **Human-in-the-Loop:** One-click triage
- ✓ **Registered Users:** Quick-login recognition

TECHNOLOGIES USED:



2026 CASE STUDY Operation Cyber Guardian: Critical Multi-Cloud Infrastructure Infiltration (2026)

Gemini 2.0 Flash AI Advantage

Evaluation Parameter	❌ 2026 Real Breach (Legacy Disconnected SOC)	✅ Present Unified Platform (With Gemini 2.0 AI)
Detection Speed (MTTD)	Undetected for >180 days across hybrid cloud & Linux servers.	Automated detection in <5 minutes via ECS 1.12 normalizer.
Attack Correlation	Disconnected tools failed to trace lateral movement across hypervisors.	Auto-maps MITRE T1078 (Credentials) & T1021 (Lateral Movement).
Forensic Audit Trail	Attackers deleted Linux syslog files, destroying evidence trails.	SHA-256 cryptographic chain immediately flags log tampering.
Analyst Response (MTTR)	Analysts overwhelmed by 100k+ alerts; manual triage took months.	Gemini 2.0 Flash provides mitigation script; resolved in 4 hours.

Live Project Links



Frontend: [unified-threat-platform.vercel.app](#)



GitHub: [github.com/Pravallika2025/unified-threat-platform](#)



Gemini 2.0 Flash Key →



Login: [admin@threatplatform.dev](#) / Admin@12345